

Anisakis simplex allergy after eating chicken meat.



BACKGROUND: Allergic reactions to food can be produced by contaminants that induce sensitization. Among these, *Anisakis simplex* can cause seafood infestation, and allergic symptoms (urticaria-angioedema, anaphylaxis, and asthma) can follow the eating or handling of affected fish. Although seafood is the principal source of human infections by this parasite, we have found allergic symptoms in 8 patients previously diagnosed as having *A simplex* sensitization after they ate chicken meat. Chicken feed usually has a high proportion of fishmeal, which might possibly be contaminated by this nematode. **OBJECTIVE:** The aim of our study was to determine whether parasite proteins present in chicken meat could be responsible for the symptoms reported by these subjects. **METHODS:** We carried out in vivo tests (prick, bronchial challenge, and double-blind placebo-controlled challenge with meat chicken) in these 8 patients. We performed immunoblotting using the sera from the 8 patients and controls in order to detect *A simplex* sensitization. We also investigated the presence of *A simplex* proteins in sera from chickens fed with fishmeal and in other sera from chickens fed only with cereals. We excluded sensitization to other chicken nematodes by serologic methods. **RESULTS:** All 8 patients presented positive prick and challenges to *A simplex*. When we used serum from chickens fed with fishmeal as the antigen in blotting, patients 3, 4, 5, 6, 7, and 8 recognized a band of 16 kd, also obtained when using pools of fish-shellfish and *A simplex* larva. No detection was observed with sera from chickens fed with only cereals. **CONCLUSION:** We provide evidence, based on in vivo and in vitro tests, that subjects highly sensitized to *A simplex* can detect the presence of *Anisakis* species allergens in chicken meat.